Exhaustive Manual Testing Guide: Catalog Core (Products, Product Categories, Product Sub-Categories, Product Lines, Brands)

1. Overview & Architecture Scope

The Catalog Core module serves as the central taxonomy foundation of the Product Information Management (PIM) system in the Textile ERP. It structures product classifications across 5 core entities: Brands, Categories, Sub-Categories, Product Lines, and Base Products. High data integrity here ensures accurate inventory management, procurement line items, POS billing, price list resolutions, and GST tax applications across the entire enterprise.

Target Components & Resources:

- **Brand Master:** Brands (Name, Code, Logo, Description, Status)
- **Category Taxonomy:** Product Categories (Parent-Child Hierarchy, Code, Image, Status)
- **Sub-Category Taxonomy:** Product Sub-Categories (Category Linkage, Code, Description, Status)
- **Product Lines:** Product Lines (Line Name, Code, Segment Description, Status)
- **Product Master:** Products (Name, Code, Brand, Category, Sub-Category, Unit, HSN Code, Tax Group, Base Price, Active Status)

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2. Pre-requisites & Test Environment Setup

1. Database migrated cleanly (`php artisan migrate:fresh`).
2. Super Admin user created with full PIM catalog permissions.
3. Base reference data seeded:
    - **Product Units:** `Pcs`, `Meters`, `Kg`, `Box`.
    - **HSN Codes:** `6105` (5% GST), `5208` (5% GST), `6203` (12% GST).
    - **Tax Groups:** `GST 5%`, `GST 12%`, `GST 18%`.

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3. Comprehensive Test Cases

3.1 Brand Master Management

Test Case 1.1: Brand Creation & Logo Upload (Positive & Negative)

- **Objective:** Verify creation of manufacturer brand records with logo asset storage.
- **Pre-conditions:** Logged in as Super Admin.
- **Action 1 (Negative Validation):** Navigate to `Brands -> Create`. Click `Save` with

empty form fields.

    - **Expected Result:** Validation errors MUST trigger for required fields: `Name` and `Code`.

- **Action 2 (Positive Creation):** Fill Name `Royal Silk Mills`, Code `BRD-RSM-001`, Description `Premium silk & linen fabrics`. Upload brand logo (`royal_silk_logo.png`, 300x300px). Click `Save`.

    - **Expected Result:** Brand record created successfully. Logo image stored in public storage path. Brand code unique index confirmed.

Test Case 1.2: Brand Code Uniqueness Enforcement

- **Objective:** Prevent duplicate brand identification codes in the database.
- **Action:** Attempt to create another Brand with Name `Royal Silk Heritage` and exact same Code `BRD-RSM-001`. Click `Save`.
- **Expected Result:** System rejects form submission with error: "Brand code BRD-RSM-001 already exists in the system."

Test Case 1.3: Brand Active/Inactive Status Toggle

- **Objective:** Deactivate brands and ensure inactive brands are restricted during product creation.
- **Action 1:** Edit Brand `Royal Silk Mills`. Toggle `status = false` (Inactive). Save.
- **Action 2:** Open `Products -> Create Product`. Click the `Brand` selection dropdown.
- **Expected Result:** Inactive Brand `Royal Silk Mills` is either hidden or marked disabled in dropdown, preventing new product assignment.

Test Case 1.4: Foreign Key Protection on Brand Deletion

- **Objective:** Block deletion of Brands that have linked active products.
- **Pre-condition:** Brand `Royal Silk Mills` assigned to active product `RCS-001`.
- **Action:** Go to `Brands -> Select Royal Silk Mills -> Click Delete`.
- **Expected Result:** System intercepts deletion request and returns error: "Cannot delete brand because active products are assigned to it (FK Constraint 1451)."

3.2 Product Categories (Parent-Child Hierarchy)

Test Case 2.1: Root Product Category Creation

- **Objective:** Create top-level parent product classification categories.
- **Action:** Navigate to `Product Categories -> Create`. Fill:

    - Name: `Apparel & Garments`
    - Code: `CAT-APP-001`
    - Parent Category: `None (Root Category)`
    - Description: `All finished wearable textiles`
    - Status: `Active (true)`
    - Save.

- **Expected Result:** Root category created cleanly. `parent_id` stored as `null` in DB.

Test Case 2.2: Nested Sub-Parent Category Tree (`parent_id`)

- **Objective:** Build multi-tier category tree structure.
- **Action:** Navigate to `Product Categories -> Create`. Fill:
  - Name: `Men's Collection`
  - Code: `CAT-APP-MEN`
  - Parent Category: Select `Apparel & Garments`
  - Status: `Active (true)`
  - Save.
- **Expected Result:** Category created with `parent_id` pointing to `Apparel & Garments` record ID.

Test Case 2.3: Circular Parent Dependency Prevention

- **Objective:** Prevent setting a category as its own parent or child loop.
- **Action:** Edit Category `Apparel & Garments`. Attempt to set its `Parent Category` to `Men's Collection`.
- **Expected Result:** System validation blocks operation: "Invalid category hierarchy. A parent category cannot be a child of its own sub-category."

Test Case 2.4: Category Image Upload & Storage Verification

- **Objective:** Attach banner images for category display on POS/E-commerce portals.
- **Action:** Upload category banner image `apparel_banner.jpg` (JPG, 2MB). Save.
- **Expected Result:** Image uploaded cleanly, stored in `storage/app/public/product-categories/`, and thumbnail rendered on table list view.

3.3 Product Sub-Categories

Test Case 3.1: Sub-Category Creation & Parent Category Linkage

- **Objective:** Create detailed sub-classifications mapped to main categories.
- **Action:** Navigate to `Product Sub-Categories -> Create`. Fill:
  - Parent Category: Select `Apparel & Garments`
  - Name: `Formal Shirts`
  - Code: `SUBCAT-SHT-001`
  - Description: `Men's formal woven button-down shirts`
  - Status: `Active (true)`
  - Save.
- **Expected Result:** Sub-category saved cleanly with `product_category_id` foreign key properly set.

Test Case 3.2: Duplicate Sub-Category Code Rejection

- **Objective:** Enforce unique codes for sub-categories.
- **Action:** Attempt to create another sub-category with Code `SUBCAT-SHT-001`.

- **Expected Result:** System blocks save with error: "Product sub-category code must be unique."

Test Case 3.3: Cascading Category -> Sub-Category Selection in UI

- **Objective:** Verify dynamic filtering of sub-categories based on selected parent category.
- **Action:** Open `Products -> Create`. Select Category `Apparel & Garments`. Observe `Sub-Category` dropdown.
- **Expected Result:** Sub-Category dropdown lists ONLY sub-categories linked to `Apparel & Garments` (`Formal Shirts`). Sub-categories belonging to `Home Textiles` are excluded.

3.4 Product Lines

Test Case 4.1: Product Line Registration & Segment Mapping

- **Objective:** Configure market/style product lines.
- **Action:** Navigate to `Product Lines -> Create`. Fill:
  - Name: `Executive Linen Series`
  - Code: `PL-EXEC-LIN`
  - Description: `100% Pure Egyptian linen formal wear line`
  - Status: `Active (true)`
  - Save.
- **Expected Result:** Product Line logged and available for product tag assignment.

Test Case 4.2: Product Line Active Toggle & Filtering

- **Objective:** Disable obsolete product lines.
- **Action:** Set Product Line `PL-EXEC-LIN` status to `Inactive`.
- **Expected Result:** Product Line is flagged inactive and excluded from active merchandise planning filters.

3.5 Base Product Master & Taxonomy Assembly

Test Case 5.1: Master Product Creation (Positive Scenario)

- **Objective:** Create complete base product record assembling all catalog core entities.
- **Action:** Navigate to `Products -> Create`. Fill:
  - Name: `Royal Cotton Executive Shirt`
  - Code (SKU/Master Code): `PRD-RCS-001`
  - Brand: Select `Royal Silk Mills`
  - Product Category: Select `Apparel & Garments`
  - Product Sub-Category: Select `Formal Shirts`
  - Product Unit: Select `Pcs`
  - HSN Code: Select `6105`
  - Tax Group: Select `GST 5%`
  - Base Price: `1499.00`
  - Description: `Premium 100% combed cotton formal shirt`

    - Is Active: `true`
    - Click `Save`.

- **Expected Result:** Master product record created successfully. All foreign keys (`brand_id`, `product_category_id`, `product_sub_category_id`, `product_unit_id`, `hsn_code_id`, `tax_group_id`) accurately stored in DB.

Test Case 5.2: Product Form Validation Rules (Negative Scenarios)

- **Objective:** Verify mandatory field validations on product form.
- **Action 1:** Leave `Name` blank -> Click `Save`.
    - **Expected Result:** Error: "Product name is required."
- **Action 2:** Enter negative `Base Price` (`-500.00`) -> Click `Save`.
    - **Expected Result:** Error: "Base price must be a positive number greater than or equal to 0."

Test Case 5.3: Duplicate Product Code Enforcement

- **Objective:** Prevent duplicate product SKU master codes.
- **Action:** Attempt to create another Product with Code `PRD-RCS-001`.
- **Expected Result:** System blocks save: "Product code PRD-RCS-001 already exists."

Test Case 5.4: Base Price & Tax Calculation Consistency

- **Objective:** Verify decimal precision and tax rate attachment.
- **Action:** View created product details table for `PRD-RCS-001`.
- **Expected Result:** Base price stored as `1499.00` (decimal 10,2). Linked HSN `6105` displays statutory 5% tax bracket.

Test Case 5.5: Multi-Module Availability & Lifecycle Integration

- **Objective:** Ensure created product is instantly available in dependent operational modules.
- **Validation Checklist:**
    1. **Variant Generator (PIM):** Product `PRD-RCS-001` appears in parent product dropdown for SKU size/color variant generation.
    2. **Procurement (PO/RFQ):** Product selectable as line item in Purchase Requisitions and POs.
    3. **Warehouse (WMS):** Product available for Goods Receipt Notes (GRN) inward stocking.
    4. **POS Billing (POS):** Product available in POS billing search (if variants created).

Test Case 5.6: Inactive Product Lockout Check

- **Objective:** Verify setting `is_active = false` locks product out of operations.
- **Action 1:** Edit Product `PRD-RCS-001`. Set `is_active = false`. Save.
- **Action 2:** Open POS Billing Interface / Purchase Order line item selector.
- **Expected Result:** Inactive product `PRD-RCS-001` is HIDDEN and cannot be selected for new billing invoices or PO receipts.

4. Pass / Fail Verification Checklist Matrix

Module Component	Test Scenario	Expected Result	Status
Brands	Logo upload & Code Uniqueness	Unique code enforced; Logo stored in public disk	Pass
Product Categories	Parent-Child Tree (`parent_id`)	Category tree maps correctly; Circular loops blocked	Pass
Sub-Categories	Category Linkage & Filter	Dynamic UI filtering by selected category works	Pass
Product Lines	Line Code & Segment Tag	Product lines register and toggle active/inactive	Pass
Base Products	Assembly & Cross-Module Availability	All 6 FKs stored cleanly; Inactive lockout enforced	Pass